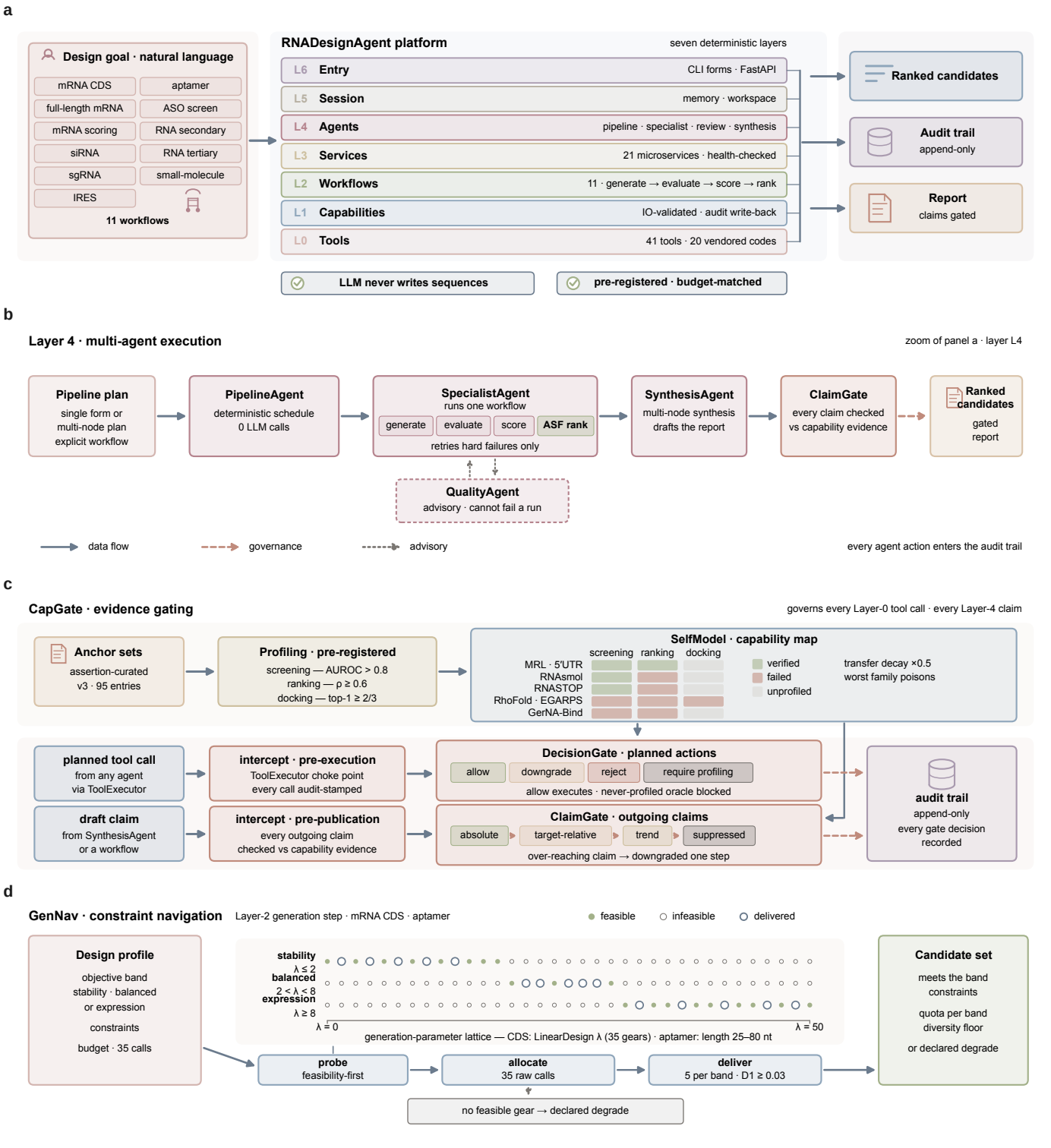

a
Design goal · natural language
mRNA CDS
aptamer
full-length mRNA
ASO screen
mRNA scoring
RNA secondary
siRNA
RNA tertiary
sgRNA
small-molecule
IRES
11 workflows
RNADesignAgent platform
seven deterministic layers
L6 Entry
CLI forms · FastAPI
L5 Session
memory · workspace
L4 Agents
pipeline · specialist · review · synthesis
L3 Services
21 microservices · health-checked
L2 Workflows
11 · generate → evaluate → score → rank
L1 Capabilities
IO-validated · audit write-back
L0 Tools
41 tools · 20 vendored codes
Ranked candidates
Audit trail
append-only
Report
claims gated
LLM never writes sequences
pre-registered · budget-matched
b
Layer 4 · multi-agent execution
zoom of panel a · layer L4
Pipeline plan
single form or multi-node plan explicit workflow
PipelineAgent
deterministic schedule
0 LLM calls
SpecialistAgent
runs one workflow
generate
evaluate
score
ASF rank
retries hard failures only
QualityAgent
advisory · cannot fail a run
SynthesisAgent
multi-node synthesis drafts the report
ClaimGate
every claim checked vs capability evidence
Ranked candidates
gated report
data flow
governance
advisory
every agent action enters the audit trail
c
CapGate · evidence gating
governs every Layer-0 tool call · every Layer-4 claim
Anchor sets
assertion-curated
v3 · 95 entries
Profiling · pre-registered
screening — AUROC > 0.8
ranking — ρ ≥ 0.6
docking — top-1 ≥ 2/3
SelfModel · capability map
screening
ranking
docking
MRL · 5'UTR
RNAsmol
RNASTOP
RhoFold · EGARPS
GerNA-Bind
verified
failed
unprofiled
transfer decay ×0.5
worst family poisons
planned tool call
from any agent via ToolExecutor
intercept · pre-execution
ToolExecutor choke point
every call audit-stamped
DecisionGate · planned actions
allow
downgrade
reject
require profiling
allow executes · never-profiled oracle blocked
draft claim
from SynthesisAgent or a workflow
intercept · pre-publication
every outgoing claim
checked vs capability evidence
ClaimGate · outgoing claims
absolute
target-relative
trend
suppressed
over-reaching claim → downgraded one step
audit trail
append-only
every gate decision recorded
d
GenNav · constraint navigation
Layer-2 generation step · mRNA CDS · aptamer
feasible
infeasible
delivered
Design profile
objective band
stability · balanced
or expression
constraints
budget · 35 calls
stability
λ ≤ 2
balanced
2 < λ < 8
expression
λ ≥ 8
λ = 0
λ = 50
generation-parameter lattice — CDS: LinearDesign λ (35 gears) · aptamer: length 25–80 nt
Candidate set
meets the band constraints
quota per band
diversity floor
or declared degrade
probe
feasibility-first
allocate
35 raw calls
deliver
5 per band · D1 ≥ 0.03
no feasible gear → declared degrade